

HDOC - 37

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Batch	15

Question

Write a menu-driven C program to perform the following operations using separate functions:

1. Convert a lowercase string to uppercase without using built-in functions.
2. Reverse a string.
3. Check if a string is a palindrome.
4. Exit.

Program

```
#include <stdio.h>

int length(char str[]) {
    int i = 0;
    while (str[i] != '\0')
        i++;
    return i;
}

void toUpperCase(char str[]) {
    int i = 0;
    while (str[i] != '\0') {
        if (str[i] >= 'a' && str[i] <= 'z')
            str[i] = str[i] - 32;
        i++;
    }
    printf("Uppercase string: %s\n", str);
}

void reverseString(char str[]) {
    int i, n = length(str);
    printf("Reversed string: ");
    for (i = n - 1; i >= 0; i--)
        printf("%c", str[i]);
    printf("\n");
}

void checkPalindrome(char str[]) {
    int i, n = length(str), flag = 1;

    for (i = 0; i < n / 2; i++) {
        if (str[i] != str[n - i - 1]) {
            flag = 0;
            break;
        }
    }

    if (flag)
        printf("The string is a palindrome.\n");
    else
        printf("The string is not a palindrome.\n");
}

int main() {
    char str[100];
    int choice, i = 0;

    printf("Enter a lowercase string: ");
    fgets(str, sizeof(str), stdin);

    while (str[i] != '\0') {
        if (str[i] == '\n') {
            str[i] = '\0';
            break;
        }
        i++;
    }

    do {
        printf("\n--- MENU ---\n");
        printf("1. Convert lowercase to uppercase\n");
        printf("2. Reverse the string\n");
        printf("3. Check palindrome\n");
    }
```

```

printf("4. Exit\n");
printf("Enter your choice: ");
scanf("%d", &choice);

switch (choice) {
case 1:
    toUpperCase(str);
    break;
case 2:
    reverseString(str);
    break;
case 3:
    checkPalindrome(str);
    break;
case 4:
    printf("Exiting program...\n");
    break;
default:
    printf("Invalid choice\n");
}
} while (choice != 4);

return 0;
}

```

Sample Output

Enter a lowercase string: madam

--- MENU ---

1. Convert lowercase to uppercase
2. Reverse the string
3. Check palindrome
4. Exit

Enter your choice: 2

Reversed string: madam

Enter your choice: 3

The string is a palindrome.

Enter your choice: 1

Uppercase string: MADAM

Enter your choice: 4

Exiting program...

Result: The program successfully converts a lowercase string to uppercase without a built-in conversion function, reverses the string, and checks whether it is a palindrome using separate functions.